Cube Roots

The cube root is the reverse of cubing. Learn the symbol $\sqrt[3]{\ }$, and the prime-factorisation method of grouping primes in threes and taking one from each group.

What is a cube root?

The idea: Cubing and taking a cube root are **opposite** operations. If **$y = x^3$**, then **x is the cube root of y**. The symbol is $\sqrt[3]{\ }$.

$$2^3 = 8 \iff \sqrt[3]{8} = 2 \quad | \quad \sqrt[3]{27} = 3, \quad \sqrt[3]{1000} = 10$$

Method — Cube root by prime factorisation

Prime factorise the number, group identical primes in **threes** (triples), and take **one factor from each triple**. Their product is the cube root.

Why threes? Each prime factor of a number appears **three times** in the prime factorisation of its cube. Grouping in triples simply undoes that.

Number	Prime factorisation	Cube root
64	(2×2×2)(2×2×2)	4
729	(3×3×3)(3×3×3)	9
3375	(3×3×3)(5×5×5)	15

Prerequisite refresher: A "triple" is a group of three identical factors, e.g. (5 × 5 × 5). One taken from the triple is a single 5.

Where marks slip away

Trap 1 — Pairing instead of tripling. Cube roots group in **threes**. Pairing is for square roots.

Trap 2 — Taking the whole triple. From (5 × 5 × 5) take **one** 5, not 125.

Trap 3 — Mixing up symbols. $\sqrt{\ }$ means square root; $\sqrt[3]{\ }$ means cube root. Don't write $\sqrt{\ }$ when you mean cube root.

Slip 4 — Leftover factors. If a prime can't form a complete triple, the number is not a perfect cube and has no whole-number cube root.

Model answers — write them like this

Example A. Find $\sqrt[3]{512}$ by prime factorisation.

Step 1: $512 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2$ ($= 2^9$).

Step 2: Group in threes: $(2 \times 2 \times 2)(2 \times 2 \times 2)(2 \times 2 \times 2)$.

Step 3: Take one from each triple: $2 \times 2 \times 2 = 8$.

$\boxed{\therefore\ \sqrt[3]{512} = 8}$

Example B. Find $\sqrt[3]{2197}$ by prime factorisation.

Step 1: $2197 = 13 \times 13 \times 13$.

Step 2: One complete triple: $(13 \times 13 \times 13)$.

Step 3: Take one factor: 13.

$\boxed{\therefore\ \sqrt[3]{2197} = 13}$

Example C. Find $\sqrt[3]{3375}$.

Step 1: $3375 = 3 \times 3 \times 3 \times 5 \times 5 \times 5$.

Step 2: Triples: $(3 \times 3 \times 3)(5 \times 5 \times 5)$.

Step 3: One from each: $3 \times 5 = 15$.

$\boxed{\therefore\ \sqrt[3]{3375} = 15}$

Example D. Find $\sqrt[3]{1728}$ by prime factorisation.

Step 1: $1728 = 2^6 \times 3^3 = (2 \times 2 \times 2)(2 \times 2 \times 2)(3 \times 3 \times 3)$.

Step 2: Take one from each triple: $2 \times 2 \times 3 = 12$.

$\boxed{\therefore\ \sqrt[3]{1728} = 12}$

Next: try the worksheet, then check the answer key.